

Fisher/Cramér-Rao Bounds for Kinematic and Phase-Shear Signal Extraction

Full-covariance θ -CRB vs. real pixel-likelihood fits; propagation to β ; the `best/moments/null` estimator spectrum; and a fresh CRB for the phase-shear readout

Practical usage. This note is the theory/derivation reference. To actually *run* any of this against a new wavefront or sequence – not just the three datasets validated here – see README.md §4d (`non_phase_shear/analysis/crb_vs_empirical.py` for the kinematic pipeline, `python-scripts/phase_sh` for phase-shear), which take the same `<dataset>/--label/ --psmap_tag` arguments as every other pipeline stage.

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1 Motivation

Looking at 2D residual plots of $\hat{\theta}_{\text{best}} - \theta_{\text{true}}$ for the `confocal_random_zernike` (aberrated-wavefront) dataset, the scatter looks Gaussian. That raises the obvious question: can a Fisher-information / Cramér-Rao bound predict that spread analytically, rather than only measuring it empirically?

This was tried once before and got stuck. From `notes/joint_profile_inference.tex` (a sibling investigation, §2026-07-20): a Laplace approximation using the *diagonal* of a local finite-difference Hessian at θ 's MAP was found **66–220× tighter** than the true empirical scatter across 50 shots. That work diagnosed why, and flagged two things that were never actually done:

1. θ 's failure mode is a *ridge* – a continuous, weakly-identified direction (near-zero Fisher eigenvalue), not discrete multimodality like ϕ 's. A ridge is compatible with local Fisher/CRB validity *if* it is close to linear/quadratic near the truth – but whether it actually is was flagged as unchecked (Future Work item 0a in that file) and never verified.
2. The failed attempt used a **diagonal-only** Hessian, discarding exactly the off-diagonal correlations a ridge produces. `crb_signal.py` already has `build_joint_fisher()` (full 17×17 joint Fisher over $(\phi, \theta_{z0}, \theta_{z100})$, pixel-likelihood mode, all cross terms) and `full_per_shot_covariance()` (inverts the *full* matrix) – but the latter was dead code, never called from that module's own driver. Nobody had tried the full-covariance version.

This note finishes that specific, previously-flagged, previously-skipped check, reusing the existing (but previously unwired) machinery rather than re-deriving anything. Scope, by explicit direction: **one dataset** (the aberrated `confocal_random_zernike` case), matching CRB to empirical data. A Gaussian-beam-vs-aberrated comparison, and the downstream connection to the β (signal) CRB, are deliberately deferred.

2 Setup

Dataset: `data/R20_N200_A1000000_...psmapconfocal_random_zernike` (20 runs $\times$ 200 shots, 10^6 atoms/shot, aberrated wavefront via `phase_space_grids.py`'s interpolated-PSMAP path). Cloud prior: independent Gaussian on each of the 8 kinematic components (position, velocity, position spread, velocity spread, in x and y), $\sigma = 10 \mu\text{m}$ (or the matching velocity unit) for all 8. Evaluator geometry: **exactly** what `non_phase_shear/analysis/generate_kinematic_estimates.py`'s **best** (pixel-likelihood) fit uses – `bins=32`, tight half-range $1.54 \times 10^{-3} \text{ m}$, `pixel_n_gh=8` – so the CRB is evaluated on the identical pixel/bin configuration that produced the empirical numbers it is compared against. Using a different geometry would make the comparison apples-to-oranges before it starts.

All CRB quantities below are evaluated at the *true* $(\phi, \theta_{z0}, \theta_{z100})$ for each shot – the standard Cramér-Rao convention, and also convenient here since it needs no extra optimizer run (the empirical $\hat{\theta}_{\text{best}}$ values already come from a separate, already-run fit).

Implementation: `non_phase_shear/analysis/crb_vs_empirical.py` (originally written as a standalone `theta_crb_vs_empirical.py`, since folded into this one script – see the note at the end of §10), importing `crb_signal.py`'s Fisher-matrix code rather than duplicating it. Runnable against any dataset/label/psmap-tag, not just this section's – see README.md §4d.

3 Step 1: Is the ridge direction actually linear/quadratic?

For 10 real shots (`run_000`), built the full 17×17 joint Fisher matrix H at the true $(\phi, \theta_{z0}, \theta_{z100})$, found its smallest eigenvalue $\lambda_{\min}$ and eigenvector v (the ridge direction; $\sigma_{\text{ridge}} = 1/\sqrt{\lambda_{\min}}$), and profiled the *real* Poisson NLL along v at $k\sigma_{\text{ridge}}$ for $k \in \{0.5, 1, 2, 4, 8\}$, comparing to the Fisher-implied quadratic $\frac{1}{2}\lambda_{\min}(k\sigma_{\text{ridge}})^2$.

Result: not cleanly quadratic. The ratio (actual ΔNLL / quadratic ΔNLL) at $k = 0.5$ – 1 (the scale that actually matters – real fluctuations live near 1σ , not 8σ) ranged from -2.9 to $+4.3$ across the 10 shots – including several shots where the sign itself flips (the true NLL is briefly *lower*, not higher, than at the anchor point, in the direction the Fisher matrix calls “increasing”). The ratio does converge toward 1 at larger k (e.g. 0.80 – 1.23 at $k = 8$), consistent with the quadratic approximation eventually dominating far from the anchor, but that is not the regime real shot noise samples.

This is a genuine, unresolved finding, not swept under the rug: the single worst-conditioned direction is *not* well-approximated by the local quadratic at the scale that matters. And yet – see §4 – the *aggregate*, full 8-dimensional marginal covariance built from the same local Fisher matrix matches the empirical scatter well. The most likely reconciliation: v (found by minimizing over the full 17-dimensional $(\phi, \theta_{z0}, \theta_{z100})$ space) is a mixed direction, not aligned with any single physical θ component’s own marginal-variance direction, so a 2 – $4\times$ local mis-estimate of curvature along v does not necessarily translate into a comparable error in any individual component’s marginal variance after the Schur-complement/inversion mixes all 17 directions together. This is a hypothesis, not something checked here – flagged in §10 as follow-up.

4 Step 2+3: Full-covariance CRB vs. empirical RMSE

For the same 10 shots, computed `full_per_shot_covariance(H)` and extracted $\sigma_{\theta_{z0}}, \sigma_{\theta_{z100}}$ (marginal std per component), aggregated as root-mean-square across shots. Compared against the empirical RMSE of $\hat{\theta}_{\text{best}} - \theta_{\text{true}}$, pooled over $z0$ and $z100$, from the 13 completed runs of the real `confocal_random_zernike` job available at the time of this check (5200 shot/z-slice points).

One data-quality issue found and handled: the raw empirical RMSE for σ_{vy} was 2.33×10^{-6} , a $3.46\times$ mismatch against the CRB’s 6.75×10^{-7} – while every other component matched to a few percent. Investigated directly rather than accepted: of 5200 points, a single shot has a σ_{vy} residual of 1.6×10^{-4} ($161 \mu\text{m}$) – roughly $70\times$ larger than the next-largest residual in that component and wildly inconsistent with the rest of the distribution (median $|\text{residual}| = 4.4 \times 10^{-7}$, matching σ_{vx} ’s median almost exactly). RMSE is quadratic in residuals, so this one likely optimizer failure (an L-BFGS-B run landing in a bad basin for that one shot – not a CRB or model problem) was enough to dominate the whole-sample RMSE by itself. Excluding it (a MAD-based robust-outlier filter, $> 20\sigma_{\text{robust}}$ in any component; 1/5200 points excluded) brings σ_{vy} ’s empirical RMSE to 6.77×10^{-7} .

Result: the full-covariance CRB matches the empirical scatter to within a few percent, across all 8 components, once one known bad fit is excluded. This directly refutes the presumption (reasonable, given the prior 66 – $220\times$ failure) that Fisher/CRB simply cannot predict θ ’s recoverability – it was never actually tried with the full covariance before now.

5 Extension: the unaberrated (Gaussian-beam) baselines

The above was scoped to the aberrated dataset only; this section repeats §4’s CRB-vs-empirical comparison (Step 2+3 only, not the ridge check) for the two already-fit, unaberrated (`CONFOCAL_FINE`)

Component	CRB (avg z_0, z_{100})	Empirical RMSE (outlier excl.)	Ratio
μ_{x0}	9.655×10^{-6}	9.816×10^{-6}	1.017
μ_{y0}	9.655×10^{-6}	9.531×10^{-6}	0.987
μ_{vx0}	2.471×10^{-6}	2.511×10^{-6}	1.016
μ_{vy0}	2.473×10^{-6}	2.437×10^{-6}	0.985
σ_x	9.500×10^{-6}	9.912×10^{-6}	1.043
σ_y	9.461×10^{-6}	9.892×10^{-6}	1.046
σ_{vx}	6.767×10^{-7}	6.861×10^{-7}	1.014
σ_{vy}	6.745×10^{-7}	6.771×10^{-7}	1.004

Table 1: Full-covariance theta CRB vs. empirical RMSE (outlier-excluded), all 8 kinematic components, confocal_random_zernike dataset. All values in metres (or the matching velocity unit). Every component agrees to within $\pm 5\%$.

baseline datasets already saved in `non_phase_shear/results/(kinematic_estimates_{1e6,1e8})_N10_shots50` – no new data generation, reusing the same `crb_vs_empirical.py` with `--label 1e6/--label 1e8` against these datasets (the standalone `theta_crb_gaussian_baseline.py` this originally used has been folded into that one script, which now auto-falls-back to reading `true_theta/n_tot` from the saved JSON when a dataset’s source images are gone, exactly as described in the 1e6 caveat below).

Data-availability caveat for 1e6. Its source dataset is no longer on disk (the only remaining 1e6-atom, 50-shots/run directory has a different cloud-width prior than what the saved `true_theta` values actually show, so it isn’t the same source). The saved JSON has `true_theta` and photon-count totals (n_{tot}) – both of which `build_joint_fisher` needs – but not the true ϕ . Since ϕ is drawn independently of θ (`phi0_mode=random`), this uses a random $\phi \sim U(0, 2\pi)$ substitute per shot instead. 1e8 has its exact source data (`data/R40_N50_A100000000..._f0.3000`, confirmed by matching `true_theta`/photon-count against its own file attrs), so its numbers use the real per-shot ϕ , no substitution.

Component	CRB (1e6)	Empirical (1e6)	Ratio
μ_{x0}	9.644×10^{-6}	1.017×10^{-5}	1.054
μ_{y0}	9.644×10^{-6}	9.676×10^{-6}	1.003
μ_{vx0}	2.500×10^{-6}	2.638×10^{-6}	1.055
μ_{vy0}	2.501×10^{-6}	2.499×10^{-6}	0.999
σ_x	9.389×10^{-6}	1.019×10^{-5}	1.086
σ_y	9.367×10^{-6}	9.990×10^{-6}	1.067
σ_{vx}	6.676×10^{-7}	7.053×10^{-7}	1.056
σ_{vy}	6.699×10^{-7}	6.909×10^{-7}	1.031

Table 2: 1e6 baseline (unaberrated, ϕ substituted – see caveat above): matches to within $\pm 9\%$, similar quality to the aberrated result.

Interpretation. 1e6 (both aberrated and unaberrated) and the aberrated case both match the CRB to a few percent. 1e8 does not – empirical scatter is systematically $1.2\text{--}1.6\times$ the CRB, worst for the width parameters. This is not swept under the rug, and a plausible explanation already exists in prior work rather than needing to be invented here: `notes/joint_profile_inference.tex` (§2026-07-20, a different optimizer but the same underlying pixel likelihood) documented that θ ’s optimisation “*genuinely struggles at 10^8* ” – even warm-started at the exact true θ , a conditional solve “wandered/ plateaued... never approaching tolerance” – attributed to the likelihood surface becoming legitimately much sharper at high photon count ($|\nabla_{\theta}\text{NLL}|$ exceeding the NLL

Component	CRB (1e8)	Empirical (1e8)	Ratio
μ_{x0}	7.616×10^{-6}	9.279×10^{-6}	1.218
μ_{y0}	7.656×10^{-6}	9.295×10^{-6}	1.214
μ_{vx0}	1.972×10^{-6}	2.402×10^{-6}	1.218
μ_{vy0}	1.982×10^{-6}	2.408×10^{-6}	1.215
σ_x	6.262×10^{-6}	9.877×10^{-6}	1.577
σ_y	6.304×10^{-6}	9.822×10^{-6}	1.558
σ_{vx}	4.297×10^{-7}	6.703×10^{-7}	1.560
σ_{vy}	4.205×10^{-7}	6.731×10^{-7}	1.601

Table 3: **1e8** baseline (unaberrated, real ϕ , no substitution): a real, non-trivial gap – means off by $\sim 21\%$, widths off by $\sim 56\text{--}60\%$. Confirmed this is not a numerical artifact: the CRB is stable under h_θ from 10^{-6} to 10^{-8} (only degrading at 10^{-9} , floating-point roundoff), and the empirical residual distribution is broadly heavy-tailed (max $\approx 4\times$ median, consistent across all 8 components) rather than dominated by one outlier shot as in §4.

itself). If `fit_theta_best`’s L-BFGS-B is similarly not fully converging at 10^8 (not checked here – see §10), that would inflate the *empirical* RMSE above the true CRB floor without the CRB itself being wrong – consistent with the direction of the observed gap (empirical $>$ CRB, not the reverse) and with 1e6 (where the same prior work reported convergence was fine) matching cleanly.

6 The full estimator spectrum: best vs. moments vs. null

Everything so far compared the CRB against **best** (the full pixel-likelihood MAP fit) only. `generate_kinematic_estimates.py` actually produces three theta estimates per shot from the same image: **best**, **moments** (a closed-form linear estimator, described below), and **null** (the prior mean, using no image information at all). Putting all three next to the CRB in one table answers a sharper question than “does the CRB match the data”: it shows *where each estimator sits between the two extremes of no information (null) and full information (the CRB, an idealised efficient estimator)* – and, since **null** and **moments** both have closed-form theoretical predictions of their own (derived below), whether an estimator’s under-performance relative to the CRB is understood *quantitatively*, not just observed as a ratio.

6.1 Empirical comparison, all three kinematic datasets

The immediate pattern – **moments** $\approx$ **null** for the four width components, across *all three* datasets regardless of aberration or photon count – is the thing that needs explaining. The next two subsections build a closed-form theoretical prediction for **null** and **moments** respectively, rather than leaving this as an empirical curiosity.

6.2 Theoretical prediction for null: trivial, and exact

null is `map_eta` with the innovation forced to zero: it always returns the prior mean m_η , using no per-shot information whatsoever. There is nothing to propagate – $\text{RMSE}(\text{null})$ is, by definition, the population spread of the true θ around the prior mean, i.e. the prior width itself. Reading straight off `map_inference.py`’s own prior constants (no new code):

$$\sigma_{\mu_{x0}} = \sigma_{\mu_{y0}} = \sigma_{\mu_{vx0}} = \sigma_{\mu_{vy0}} = \text{TAU_MU_POS} = \text{TAU_MU_VEL} = 10 \mu\text{m}.$$

For the width components the prior lives on the *variance* $V_{x0} = \sigma_x^2$, not σ_x itself: $V_{x0} \sim \mathcal{N}(\bar{\sigma}_{\text{pos}}^2, \tau_{V,\text{pos}}^2)$ with $\tau_{V,\text{pos}} = 2\bar{\sigma}_{\text{pos}} \text{SIG_SIG_POS}$ (`build_prior`’s own definition). The delta

Component	CRB	best	moments	null
<i>confocal_random_zernike (aberrated), 15 runs, 6000 pts</i>				
μ_{x0}	9.655×10^{-6}	9.778×10^{-6} (1.01 $\times$)	9.777×10^{-6} (1.01 $\times$)	1.007×10^{-5} (1.04 $\times$)
μ_{y0}	9.655×10^{-6}	9.566×10^{-6} (0.99 $\times$)	9.563×10^{-6} (0.99 $\times$)	9.809×10^{-6} (1.02 $\times$)
μ_{vx0}	2.471×10^{-6}	2.504×10^{-6} (1.01 $\times$)	2.663×10^{-6} (1.08 $\times$)	1.007×10^{-5} (4.07 $\times$)
μ_{vy0}	2.473×10^{-6}	2.447×10^{-6} (0.99 $\times$)	2.626×10^{-6} (1.06 $\times$)	1.007×10^{-5} (4.07 $\times$)
σ_x	9.500×10^{-6}	9.988×10^{-6} (1.05 $\times$)	1.000×10^{-5} (1.05 $\times$)	1.000×10^{-5} (1.05 $\times$)
σ_y	9.461×10^{-6}	9.973×10^{-6} (1.05 $\times$)	9.997×10^{-6} (1.06 $\times$)	1.000×10^{-5} (1.06 $\times$)
σ_{vx}	6.767×10^{-7}	6.913×10^{-7} (1.02 $\times$)	9.151×10^{-6} (13.5 $\times$)	1.024×10^{-5} (15.1 $\times$)
σ_{vy}	6.745×10^{-7}	6.821×10^{-7} (1.01 $\times$)	8.984×10^{-6} (13.3 $\times$)	1.005×10^{-5} (14.9 $\times$)
<i>1e8 (Gaussian beam), 10 runs, 1000 pts</i>				
μ_{x0}	7.616×10^{-6}	9.279×10^{-6} (1.22 $\times$)	9.276×10^{-6} (1.22 $\times$)	9.545×10^{-6} (1.25 $\times$)
μ_{y0}	7.656×10^{-6}	9.295×10^{-6} (1.21 $\times$)	9.298×10^{-6} (1.21 $\times$)	9.737×10^{-6} (1.27 $\times$)
μ_{vx0}	1.972×10^{-6}	2.402×10^{-6} (1.22 $\times$)	2.481×10^{-6} (1.26 $\times$)	9.830×10^{-6} (4.99 $\times$)
μ_{vy0}	1.982×10^{-6}	2.408×10^{-6} (1.21 $\times$)	2.466×10^{-6} (1.24 $\times$)	1.028×10^{-5} (5.19 $\times$)
σ_x	6.262×10^{-6}	9.877×10^{-6} (1.58 $\times$)	9.866×10^{-6} (1.58 $\times$)	9.867×10^{-6} (1.58 $\times$)
σ_y	6.304×10^{-6}	9.822×10^{-6} (1.56 $\times$)	9.833×10^{-6} (1.56 $\times$)	9.837×10^{-6} (1.56 $\times$)
σ_{vx}	4.297×10^{-7}	6.703×10^{-7} (1.56 $\times$)	8.914×10^{-6} (20.8 $\times$)	1.001×10^{-5} (23.3 $\times$)
σ_{vy}	4.205×10^{-7}	6.731×10^{-7} (1.60 $\times$)	8.965×10^{-6} (21.3 $\times$)	1.010×10^{-5} (24.0 $\times$)
<i>1e6 (ϕ-substituted), 10 runs, 1000 pts</i>				
μ_{x0}	9.644×10^{-6}	1.017×10^{-5} (1.05 $\times$)	1.017×10^{-5} (1.05 $\times$)	1.048×10^{-5} (1.09 $\times$)
μ_{y0}	9.644×10^{-6}	9.676×10^{-6} (1.00 $\times$)	9.678×10^{-6} (1.00 $\times$)	1.003×10^{-5} (1.04 $\times$)
μ_{vx0}	2.500×10^{-6}	2.638×10^{-6} (1.06 $\times$)	2.702×10^{-6} (1.08 $\times$)	1.023×10^{-5} (4.09 $\times$)
μ_{vy0}	2.501×10^{-6}	2.499×10^{-6} (1.00 $\times$)	2.557×10^{-6} (1.02 $\times$)	9.992×10^{-6} (4.00 $\times$)
σ_x	9.389×10^{-6}	1.019×10^{-5} (1.09 $\times$)	1.019×10^{-5} (1.09 $\times$)	1.020×10^{-5} (1.09 $\times$)
σ_y	9.367×10^{-6}	9.990×10^{-6} (1.07 $\times$)	1.000×10^{-5} (1.07 $\times$)	1.000×10^{-5} (1.07 $\times$)
σ_{vx}	6.676×10^{-7}	7.053×10^{-7} (1.06 $\times$)	8.922×10^{-6} (13.4 $\times$)	1.003×10^{-5} (15.0 $\times$)
σ_{vy}	6.699×10^{-7}	6.909×10^{-7} (1.03 $\times$)	8.826×10^{-6} (13.2 $\times$)	9.919×10^{-6} (14.8 $\times$)

Table 4: RMSE per component (metres or matching velocity unit), ratio to CRB in parentheses. **best** tracks the CRB everywhere. **moments** tracks the CRB for the four position/velocity *means* but is 13–21 $\times$ worse than the CRB for the four cloud-*width* components – almost identical to **null**, i.e. for width recovery **moments** is barely better than ignoring the image. **null** is $\sim 1\times$ CRB for the means (coincidence of prior width $\approx$ CRB scale here) and 4–24 $\times$ for the velocity mean/width components (real information gap, as expected for an estimator using zero data). Source: `non_phase_shear/analysis/crb_vs_empirical.py` (README.md §4d), run separately against each of the three datasets/labels.

method for $\sigma_x = \sqrt{V_{x0}}$ gives

$$\text{std}(\sigma_x) \approx \frac{\tau_{V,\text{pos}}}{2\bar{\sigma}_{\text{pos}}} = \text{SIG_SIG_POS} = 10 \mu\text{m},$$

exactly (the factor of $2\bar{\sigma}_{\text{pos}}$ in $\tau_{V,\text{pos}}$ ’s own definition cancels the delta-method’s own $2\bar{\sigma}_{\text{pos}}$ Jacobian factor – `build_prior` was evidently written with exactly this identity in mind). So the theoretical prediction is $10 \mu\text{m}$ for *all eight* components, uniformly. Table 4 shows this matches empirical **null** RMSE to within 0–9% across all three datasets – as it must, since by construction **null** carries zero per-shot Fisher information and this is purely a statement about how the data-generating prior is parameterised.

6.3 Theoretical prediction for moments: linear Poisson-noise propagation

`theta_moments` is *exactly* linear in the raw image moments $M = [\mu_{xf}, \mu_{yf}, V_{xf}, V_{yf}]$ (`map_inference.py`’s `build_A/build_prior/precompute_kalman/map_eta`):

$$\hat{\eta}_{\text{moments}} = m_{\eta} + K(M - A m_{\eta}), \quad K = P_{\eta} A^{\top} (A P_{\eta} A^{\top})^{-1} \quad (10 \times 4, \text{ fixed}),$$

so its covariance is *exactly* $\text{Cov}(\hat{\eta}_{\text{moments}}) = K \text{Cov}(M) K^\top$ – no finite differences needed, just the measurement noise of M itself propagated through this fixed linear map.

Poisson statistics of the raw moments. The port-summed pixel row sums $n_i \sim \text{Poisson}(\lambda_i)$, $i = 1, \dots, \text{bins}$, are independent, with $N = \sum_i n_i$. This is the standard Poisson-thinning identity: conditional on N , (n_i) is *exactly* Multinomial(N, p_i), $p_i = \lambda_i / \sum \lambda$. So, conditional on the realised N (what is actually observed and used), $\mu_x = \sum n_i x_i / N$ and $V_x = \sum n_i (x_i - \mu_x)^2 / N$ behave *exactly* like the sample mean/variance of N i.i.d. draws from the pixel distribution $p(x)$. The standard (shape-agnostic) finite-sample results for a weighted mean/variance estimator are

$$\text{Var}(\mu_x) = \frac{V_x}{N}, \quad \text{Cov}(\mu_x, V_x) = \frac{\mu_3}{N}, \quad \text{Var}(V_x) = \frac{\mu_4 - V_x^2}{N},$$

where μ_3, μ_4 are the 3rd/4th central moments of $p(x)$, computed directly from each shot’s own measured marginal histogram (not assumed Gaussian – the PSF-blurred image marginal need not be exactly so). The extra scatter from N itself being random rather than fixed is second order in $1/N$ ($\text{Var}(\mu_x) = \text{E}[\text{Var}(\mu_x|N)] + \text{Var}(\text{E}[\mu_x|N])$, and the second term vanishes because $\text{E}[\mu_x|N] = \mu_{x,\text{true}}$ regardless of N) and is dropped. The x/y blocks are treated as uncorrelated (exact in the underlying cloud model, an approximation for the measured image since both marginals come from the same shot’s pixel counts, but the leading-order terms above only depend on each marginal separately).

Evaluated on 20 real shots of `confocal_random_zernike` (the `moments_theory/null_theoretical_rmse` functions in `non_phase_shear/analysis/crb_vs_empirical.py`, originally a standalone `moments_null_crb_vs_` this Poisson-only prediction is:

Component	Poisson-noise-only theory	Empirical <code>moments</code> RMSE (ratio)
μ_{x0}	4.7×10^{-8}	9.78×10^{-6} (207×)
μ_{y0}	1.8×10^{-7}	9.56×10^{-6} (53×)
μ_{vx0}	4.7×10^{-8}	2.66×10^{-6} (57×)
μ_{vy0}	1.8×10^{-7}	2.63×10^{-6} (15×)
σ_x	2.3×10^{-13}	1.00×10^{-5} (4×10^7)
σ_y	4.4×10^{-11}	1.00×10^{-5} (2×10^5)
σ_{vx}	3.3×10^{-12}	9.15×10^{-6} (3×10^6)
σ_{vy}	2.3×10^{-13}	9.0×10^{-6} (4×10^7)

Poisson counting noise is not the answer – it predicts an error 4–8 orders of magnitude below what’s actually observed. At 10^6 – 10^8 detected atoms, the raw image moments are simply too well-determined by counting statistics alone for shot noise to matter; `moments`’s real error must come from somewhere else entirely.

6.4 Where `moments`’s real error comes from: shrinkage bias and PSF blindness

A is 4×10 : four raw-moment measurements cannot determine ten η components, so K is a *regularised* (prior-shrunk) pseudo-inverse, not a true inverse. Even with a perfectly noiseless measurement $M = A \eta_{\text{true}}$ (no Poisson noise at all), the recovered $\hat{\eta} = m_\eta + KA(\eta_{\text{true}} - m_\eta) \neq \eta_{\text{true}}$ in general – only equal in directions where KA acts as the identity. This is a *bias*, not a variance, and it doesn’t shrink with more photons (unlike the Poisson term above). Computing this shrinkage bias directly, $(I - KA)(\eta_{\text{true}} - m_\eta)$, over the true θ values of the full aberrated dataset (4200 points) and converting back to θ -space:

Component	Shrinkage-bias-only prediction	Empirical <code>moments</code> RMSE	Ratio
μ_{x0}	2.56×10^{-6}	9.78×10^{-6}	3.8× (too small)
μ_{vx0}	9.71×10^{-6}	2.66×10^{-6}	3.7× (too large)
σ_x	9.09×10^{-8}	1.00×10^{-5}	110× (too small)
σ_{vx}	1.32×10^{-6}	9.15×10^{-6}	6.9× (too small)

(y-components essentially identical by symmetry, omitted.)

For the position/velocity *means*, this gets the right order of magnitude (a few $\times$) – μ_{x0} and μ_{vx0} come out swapped in scale relative to a naive ”read the diagonal of $I - KA$ ” expectation, but that’s exactly what’s expected once KA ’s off-diagonal structure is accounted for: a single scalar measurement (μ_{xf}) is trying to inform *two* unknowns (μ_{x0} and μ_{vx0}) simultaneously, so K ’s corresponding 2×2 sub-block genuinely mixes credit between them rather than acting diagonally – not a bug, just a reminder that $4 \rightarrow 10$ -dimensional shrinkage doesn’t decompose component-by-component.

For the *width* components, the shrinkage-bias-only picture is wrong by $100\times$ or more, and the reason is physical rather than statistical: A (`build_A`) models *pure ballistic expansion* – $V_{xf} = V_{x0} + 2TC_{xv0} + T^2V_{vx0}$ – with **no imaging point-spread-function term at all**. The real detected image’s second moment V_{xf} is inflated by the optical PSF on top of the true ballistic spread, by an amount this idealised A has no way to know about, so the ”exact constraint” $M = A\eta$ this whole propagation assumes is itself wrong at the scale of the PSF – a model-misspecification bias, not a statistical-shrinkage one, and much larger than either term computed above. This is consistent with everything already observed: `moments`’s width recovery tracks `null` almost exactly (Table 4) – i.e. the raw second image moment carries essentially *no* usable width information once the PSF is present – while `best` (which fits the true PSF-aware forward model via the PSMAP, not the raw moments) recovers width to the CRB. Quantifying the PSF-variance term explicitly to close this last gap numerically is listed in §10.

7 Framing: this is a single-shot comparison, and that’s intentional

θ is a fresh, independent nuisance parameter drawn from the prior *every shot* – it is not a shared, estimated-once parameter like $\beta = (A_s, A_c)$. Consequently, pooling more shots does *not* reduce the true per-shot θ RMSE (there is nothing to pool: shot i ’s θ_i tells you nothing about shot j ’s θ_j). What more shots *do* improve is the *measurement precision of the empirical RMSE estimate itself* – with 5200 points the empirical RMSE’s own relative uncertainty is $\sim 1/\sqrt{2 \times 5200} \approx 1\%$, tight enough to distinguish the observed few-percent CRB agreement from noise. This is why this comparison uses many shots (for a trustworthy *measurement*) while still being a fundamentally single-shot statement (the CRB itself is a per-shot quantity, not one that shrinks with N_{shots}) – unlike β ’s CRB, which genuinely does improve with shot count (`crb_signal.py`’s `crb_from_j/simple_scaling_crb`).

8 From θ -CRB to β -CRB

8.1 Method: propagating theta uncertainty into a phase-information number

`crb_signal.py`’s `per_shot_information` is exactly the theta-uncertainty-propagation step: it takes the same joint 17×17 Fisher matrix H used in §4 (over $\phi, \theta_{z0}, \theta_{z100}$) and Schur-complements *only* the 16 theta directions out (not ϕ), leaving a scalar effective per-shot phase information j_{eff} – literally ”how much information about the laser phase ϕ_i survives, after accounting for the fact that θ is unknown and must be estimated from the same image.” Since $\delta\phi_i(\beta) = A_s \sin(2\pi fi) + A_c \cos(2\pi fi)$ enters only through ϕ_i ’s argument at $z100$, the N -shot Fisher matrix for $\beta = (A_s, A_c)$ is exactly $F_\beta = \sum_i j_i x_i x_i^\top$, $x_i = [\sin(2\pi fi), \cos(2\pi fi)]$ (`crb_from_j`), and $\text{CRB}(\beta) = F_\beta^{-1}$.

8.2 Converting to rad/ $\sqrt{\text{shot}}$ and rad/ $\sqrt{\text{Hz}}$

Two useful normalisations, both already implemented:

- **rad/ $\sqrt{\text{shot}}$** (a shot-count-independent number). When shots densely sample many signal periods ($\langle x_i x_i^\top \rangle \rightarrow \frac{1}{2}I_2$, true here: $f = 0.3 \text{ cyc/shot}$ over 50-200 shots is 15-60 periods), $F_\beta \approx (N j_{\text{bar}}/2)I_2$, so $\sigma_{A_s} = \sigma_{A_c} \approx \sqrt{2/(N j_{\text{bar}})} = \sigma_{1\text{shot}}/\sqrt{N}$ (`simple_scaling_crb`). $\sigma_{1\text{shot}} = \sqrt{2/j_{\text{bar}}}$ is what "rad/ $\sqrt{\text{shot}}$ " means: it is the number such that the CRB at any shot count N is $\sigma_{1\text{shot}}/\sqrt{N}$ – report this number once, and it answers "what precision at N shots" for any N without recomputing anything.
- **rad/ $\sqrt{\text{Hz}}$** (an amplitude spectral density, the standard sensitivity unit in precision metrology). This requires one more piece of information this codebase does not define anywhere: the real shot *repetition rate* f_{rep} (shots/second) – every frequency in this codebase (`--signal_freq`) is in cycles/*shot*, not Hz, so there is no absolute time axis without it. Given f_{rep} , standard white-noise-ASD conversion applies:

$$\text{ASD [rad}/\sqrt{\text{Hz}}] = \sigma_{1\text{shot}} [\text{rad}] \times \sqrt{T_{\text{rep}}} = \sigma_{1\text{shot}}/\sqrt{f_{\text{rep}}}, \quad T_{\text{rep}} = 1/f_{\text{rep}}.$$

(Same logic as converting a per-sample noise std to a PSD/ASD for any white, sample-limited process: divide by $\sqrt{\text{sample rate}}$. Valid here because each shot's θ_i, ϕ_i are drawn independently – no shot-to-shot correlation to worry about.) $\sigma_{1\text{shot}}$ is reported below; multiply by your own f_{rep} whenever you have one.

Implementation: the beta-CRB block of `non_phase_shear/analysis/crb-vs-empirical.py` (originally a standalone `beta_crb-vs-empirical.py`). Uses a representative sample of shots (not all N) to estimate j_{bar} (checked: relative shot-to-shot spread of j is $\sim 9\%$ for **1e6**, tight enough that treating $j_i \approx j_{\text{bar}}$ for all N shots when building F_β via the real x_i pattern is a good approximation, while avoiding an $O(N)$ -shot Fisher-matrix cost).

8.3 Result: 1e6 (Gaussian beam)

	$\sigma_{1\text{shot}} [\text{rad}]$	CRB($N=50$)	Empirical ($N=50$, best)
A_s	1.14×10^{-3}	1.62×10^{-4}	4.95×10^{-4}
A_c		1.62×10^{-4}	3.21×10^{-4}

Table 5: $j_{\text{bar}} = 1.53 \times 10^6$ (rel. shot-to-shot spread 9%), 30 sample shots. Ratio (empirical/CRB): $3.05 \times (A_s)$, $1.99 \times (A_c)$.

Unlike θ 's own CRB (which matched to $\pm 5\%$), β 's CRB under-predicts the empirical scatter by a real, substantial factor. Checked and ruled out one concrete hypothesis before accepting this: rebuilt the CRB using *exactly* `beta_fits_from_kinematics.py`'s own evaluator geometry (`bins=16`, native half-range, vs. the tight/32-bin geometry used elsewhere in this note) – j_{bar} changed by $< 0.3\%$ (1.523×10^6 vs. 1.526×10^6), so evaluator geometry is *not* the explanation.

The actual explanation is structural. `beta_fits_from_kinematics.py`'s best method is a *two-stage plug-in* estimator, not the idealised joint estimator this CRB models:

1. Fit $\hat{\theta}_{\text{best}}$ first (a point estimate – whose own scatter matches its CRB, §4).
2. Convert to `crb_signal.py`'s η representation via `theta_to_eta`, which **hardcodes two cross-terms to zero**: `[mu_x0, mu_vx0, mu_y0, mu_vy0, sx**2, 0.0, svx**2, sy**2, 0.0, svy**2]` – exactly the position-velocity covariance terms.

3. Fit β from that fixed, cross-term-stripped $\eta - \theta$'s uncertainty has already been collapsed to a point estimate and its correlation structure discarded, not properly marginalised.

This CRB assumes an efficient estimator that uses the full joint (ϕ, θ, β) information optimally (Fisher-matrix elimination is the textbook-correct way to get the marginal CRB for *an efficient* estimator). The real pipeline throws away exactly the correlation structure θ was shown to actually have (§3's ridge) partway through. A two-stage plug-in estimator falling short of the joint bound is a standard, expected effect, not a bug in either number – the CRB is correctly a lower bound; this particular real pipeline doesn't saturate it.

8.4 Result: 1e8 and the aberrated dataset (confocal_random_zernike)

Same method, same N used in the corresponding empirical `beta_fits*.json` run ($N=50$ for 1e8, $N=200$ for the aberrated dataset), plus `moments/null` for completeness.

Dataset	$\sigma_{1\text{shot}}$	CRB(N)	best	moments
1e8 ($N=50$)	1.25×10^{-4}	1.77×10^{-5}	$A_s=7.40 \times 10^{-5}$ (4.18 $\times$) $A_c=8.63 \times 10^{-5}$ (4.87 $\times$)	$A_s=5.66 \times 10^{-4}$ (32.0 $\times$) $A_c=9.36 \times 10^{-4}$ (52.9 $\times$)
confocal_random_zernike ($N=200$)	1.43×10^{-3}	1.01×10^{-4}	$A_s=2.92 \times 10^{-3}$ (28.9 $\times$) $A_c=2.24 \times 10^{-3}$ (22.2 $\times$)	$A_s=1.31 \times 10^{-3}$ (12.9 $\times$) $A_c=9.10 \times 10^{-4}$ (9.0 $\times$)

Table 6: $j_{\text{rel.std}} = 4.6\%$ (1e8), 3.9% (aberrated), 30 sample shots each; 1e8 empirical from 10 runs, aberrated from the 14 runs completed at check time. All β units rad.

Two things worth calling out beyond the already-established two-stage plug-in gap (§8.3):

The gap widens with photon count and with aberration. `best`'s ratio to the CRB grows from 2–3 $\times$ (1e6) to 4–5 $\times$ (1e8) to 22–29 $\times$ (aberrated) – tracking exactly the same progression already seen for θ 's own recovery quality (§5: 1e8's 1.2–1.6 $\times$ theta gap; §4: the aberrated ridge is a much harder, more correlated recovery problem than either Gaussian baseline). This is consistent with the two-stage mechanism (§8.3) simply compounding: the worse θ 's own point estimate and discarded correlation structure are, the more the beta fit built on top of it suffers.

Counter-intuitively, for the aberrated dataset, moments-based β actually beats best-based β (12.9–9.0 $\times$ CRB vs. 28.9–22.2 $\times$), even though `moments`' own θ recovery is uniformly worse than `best`'s (Table 4) – and even though for the two Gaussian baselines the ordering is the "expected" one (`best` beats `moments` for β , matching the θ -level ordering). Not chased down further here (flagged in §10), but the most likely explanation is again structural rather than a fluke: `best`'s particular two-stage failure mode (§8.3) throws away theta's ridge *correlation*, and the aberrated wavefront is exactly the regime where that ridge is most pronounced (§3) – so the aberrated case may be exactly where discarding it hurts `best` the most, while `moments`, despite worse marginal accuracy, has a different (still crude, but differently-structured) error that happens not to interact as destructively with the beta fit's own linear-regression step.

9 Phase-shear readout: a structurally different model

Everything above uses the "kinematic" pipeline: two-port images fit per-shot for the full 8-parameter cloud state θ , with the laser phase ϕ_i profiled out and $\beta = (A_s, A_c)$ fit across shots from the $\hat{\phi}_i$ sequence. The *phase-shear* readout (`phase_shear_fit.py`) is a completely different measurement strategy for the same underlying β : instead of resolving θ and ϕ_i separately, it fits an 11-parameter Gaussian $\times$ fringe model directly to the *ground-state* port image,

$$\text{model_image}(x, y) = \exp(\log A) \cdot \exp\left[-\frac{(x-\mu_x)^2}{2\sigma_x^2} - \frac{(y-\mu_y)^2}{2\sigma_y^2}\right] \cdot [1 + C_{\cos} \cos(\cdot) + C_{\sin} \sin(\cdot) + \dots],$$

independently at $z=0$ and $z=100$, via a plain (no-prior) Poisson MLE. The fitted fringe phase $\phi_0 = \arctan 2(C_{\sin}, C_{\cos})$ at each z gives a *differential* phase $\Delta\phi = \phi_0^{z100} - \phi_0^{z0}$, which cancels the (unknown, shot-to-shot random) laser phase common to both images and is the observable this pipeline actually uses to infer β . This section asks the same question as before – does a CRB predict the empirical scatter – for this structurally different model, built entirely fresh (not reusing `crb_signal.py`'s PSMAP machinery, since there is no PSMAP forward model here, just a closed-form fringe fit).

9.1 Fisher matrix and $\text{CRB}(\Delta\phi)$

At a fitted parameter point p (11-vector), the Poisson Fisher matrix for `model_image` is built by finite differences directly on the $n_{\text{bins}} \times n_{\text{bins}}$ pixel grid (cheap: no PSMAP lookup, just a closed-form elementwise expression),

$$H(p) = J^\top W J, \quad J_k = \frac{\partial \text{model_image}}{\partial p_k}, \quad W = \text{diag}(1/\mu(p)),$$

inverted with *no* prior term (`phase_shear_fit.py`'s objective has none), then delta-methoded through $\phi_0 = \arctan 2(C_{\sin}, C_{\cos})$ using the 2×2 $(C_{\cos}, C_{\sin})$ sub-covariance:

$$\text{Var}(\phi_0) = \nabla \phi_0^\top \text{Cov}(C_{\cos}, C_{\sin}) \nabla \phi_0, \quad \nabla \phi_0 = \left[\frac{-C_{\sin}}{C_{\cos}^2 + C_{\sin}^2}, \frac{C_{\cos}}{C_{\cos}^2 + C_{\sin}^2} \right].$$

ϕ_0^{z0} and ϕ_0^{z100} are fit *independently* (unlike the kinematic pipeline's shared ϕ_i), so $\text{Var}(\Delta\phi) = \text{Var}(\phi_0^{z0}) + \text{Var}(\phi_0^{z100})$, and the effective per-shot Fisher information for $\Delta\phi$ is $j_{\text{eff}} = 1/\text{Var}(\Delta\phi)$ – feeding into *exactly* the same β -CRB machinery as §8 (`crb_from_j`), since $\Delta\phi_i$ plays the same role ϕ_i did there.

9.2 rad/ $\sqrt{\text{shot}}$ for this pipeline

The conversion is identical in form to §8.2 – j_{eff} here is a differently-computed quantity (single-image delta-method information about a fringe phase, vs. the theta-marginalised Schur complement of §8.1), but once you have a per-shot j , everything downstream is the same formula: $\sigma_{\Delta\phi, 1\text{shot}} = \sqrt{1/j_{\text{eff}}}$ is the shot-count-independent number (note: no factor of 2 here, unlike §8.2's $\sqrt{2/j}$ – that factor came from β having *two* free parameters A_s, A_c sharing one scalar phase information j per shot; here we're first reporting $\Delta\phi$'s own single-parameter CRB, one number, before it separately propagates into the 2-parameter β CRB via `crb_from_j` exactly as in §8.1), and $\text{rad}/\sqrt{\text{Hz}}$ again requires the undefined shot rate f_{rep} : $\text{ASD} = \sigma_{1\text{shot}}/\sqrt{f_{\text{rep}}}$.

9.3 What "CRB vs. empirical" means here: the systematic-bias caveat

This is the key thing to understand before reading the numbers below. The CRB models *pure Poisson shot noise for a model assumed to be exactly correctly specified* – i.e. it answers "if the only source of error were photon counting statistics, and the 11-parameter fringe model were a perfect description of the image, how precisely could $\Delta\phi$ be recovered?" The empirical results in `phase_shear/paper/phase_shear_results.tex` show that the *raw* $\Delta\phi$ residual has a large **systematic** component on top of this – a "wavefront-averaging effect" that depends on the cloud's position/shape and does *not* cancel between the $z0$ and $z100$ fits – which the published analysis regresses out using either a fitted-feature correction (in-sample/out-of-sample) or an oracle (true-parameter) correction. That regression removes most, but not all, of the systematic, leaving a corrected residual that is closer to – but still not identical to – the pure noise floor the CRB models.

So there are three fundamentally different numbers in play, and comparing the CRB only to the *raw* one would be comparing apples to oranges: the **raw** residual (systematic + noise, the

actual as-built pipeline’s current performance), the **corrected** residual (fitted-feature or oracle, systematic mostly removed, the closest existing empirical proxy for ”how good could this get”), and the **CRB** (noise only, by construction the floor *under* even the corrected number, since a real regression correction is never perfectly efficient). Reporting all three side by side, rather than picking one, is the only honest way to present this comparison.

9.4 Empirical residual: a methodological trap, documented so it doesn’t recur

The exact definition of the published residual was reverse-engineered from `phase_shear/notebooks/shear mle` (via `jupyter nbconvert`) because a first, naive attempt – `wrap( $\phi_0^{z100} - \phi_0^{z0} - \Delta\phi_{\text{true}}$ )` – gave ≈ 142 mrad for `1e6`, roughly $10\times$ the published 13.84 mrad. The notebook’s actual procedure has two non-obvious pieces:

1. A **sign flip**: the ”estimate” of the injected signal is *negated* relative to the naive $\phi_0^{z100} - \phi_0^{z0}$ difference (a sign convention in how the notebook defines its estimator, not a bug – but invisible unless you read the notebook’s own code).
2. The DC/mean offset (an overall constant, expected: $\Delta\phi_{\text{true}}$ varies shot-to-shot with a nonzero mean baked into the injected signal generator) is removed via a proper *circular*-mean unwrap, *then* an ordinary (non-circular) mean-subtraction on the already-unwrapped values – not a single naive wrapped subtraction.

With the corrected formula, the reproduced number is 13.15–13.58 mrad against the published 13.84 mrad (small remaining difference presumably a different run subset, not a formula mismatch) – confirmed correct. Implementation: `python-scripts/phase_shear_crb_vs_empirical.py`’s `empirical_delta_phi_std`, with the reasoning documented in its own docstring so this exact class of bug doesn’t recur.

9.5 Results

	1e6	1e8
j_{bar} (per shot, 45 sample shots)	5.95×10^4	5.82×10^6
CRB($\Delta\phi$), per-shot std	4.10 mrad	0.41 mrad
Empirical $\Delta\phi$, raw (this note’s formula)	13.15 mrad	11.55 mrad
ratio raw/CRB	$3.21\times$	$27.9\times$
Published raw	13.84 mrad ($3.37\times$)	13.51 mrad ($32.6\times$)
Published fitted-feat. (IS)	6.65 mrad ($1.62\times$)	1.18 mrad ($2.85\times$)
Published fitted-feat. (OOS)	7.22 mrad ($1.76\times$)	1.85 mrad ($4.46\times$)
Published oracle	7.07 mrad ($1.72\times$)	1.08 mrad ($2.60\times$)
CRB(β), $N=200/N=20$ shots/run	0.41 mrad (both)	0.13 mrad (both)
Published β raw	5.35 mrad ($13.1\times$)	5.35 mrad ($40.8\times$)
Published β fitted-feat.	2.63 mrad ($6.4\times$)	0.46 mrad ($3.5\times$)
Published β oracle	2.64 mrad ($6.4\times$)	0.33 mrad ($2.5\times$)

Table 7: Ratios in parentheses are empirical/CRB. Source: `python-scripts/phase_shear_crb_vs_empirical.py`; published numbers from `phase_shear/paper/phase_shear_results.tex`.

Interpretation, directly answering ”what is this comparison saying.” Three separate, consistent findings:

1. **The raw pipeline is nowhere near its own noise floor** – $3\text{--}33\times$ off, and (crucially) the gap gets *worse*, not better, going from `1e6` to `1e8`. This is the clearest possible

signature of a systematic, non-shot-noise error: shot noise shrinks as photon count grows (the CRB itself drops $10\times$, from 4.1 to 0.41 mrad), but the raw empirical residual barely moves ($13.15 \rightarrow 11.55$ mrad) because it is dominated by the photon-count-*independent* wavefront-averaging systematic, not by counting statistics. A shrinking denominator with a flat numerator is exactly why the ratio explodes at $1e8$.

2. **Once the systematic is regressed out (fitted-feature or oracle), the residual drops to within $1.6\text{--}4.5\times$ of the CRB** – much closer, and now in the same direction/range as every other “**best**”-style CRB gap seen elsewhere in this note (§8.3, §8.4: real, structural, estimator-inefficiency gaps of a few $\times$, not systematic-bias gaps of $10\text{--}40\times$). This is the fair comparison: it says the phase-shear readout’s *achievable* precision, once a good systematic correction is applied, is a few $\times$ the fundamental shot-noise floor – the same kind of gap the kinematic pipeline’s own **best** estimator has relative to its CRB, not a qualitatively different failure.
3. **The CRB is doing its job correctly in both cases** – it was never claiming to predict the raw number (that would require modeling the wavefront-averaging systematic, which is a model-misspecification effect a Poisson-noise CRB fundamentally cannot capture, by construction). Its value is as the *lower bound against which to judge how much of the raw error is systematic vs. irreducible*: here, at $1e8$, essentially all of the raw 11.55 mrad is systematic (1.08–1.85 mrad of it survives correction, against a 0.41 mrad floor) – so future effort is far better spent improving the systematic correction (or removing its cause) than on anything noise-related.

10 Open questions / future work

- **Reconcile §3 and §4.** Check what the worst-eigenvalue direction v actually projects onto in θ -component space, for a few shots – confirm (or refute) the “mixed direction, doesn’t dominate any single marginal” hypothesis directly rather than leaving it as speculation.
- **Diagnose the 10^8 gap (§5):** check whether `fit_theta.best`’s L-BFGS-B is actually converging (`res.nit/res.success`, and/or a tighter `maxiter` / warm-start-from-truth spot check) at 10^8 atoms specifically. If convergence is the cause, this is an estimator-quality issue, not a CRB-validity issue – worth confirming directly rather than resting on the plausible-but-unverified analogy to the joint-solve’s documented 10^8 struggles.
- **Exact (not ϕ -substituted) $1e6$ baseline** and a dedicated apples-to-apples aberrated-vs-Gaussian comparison at matched N : the $1e6/1e8$ baselines used here are convenient (already fit, no new compute) but neither is parameter-matched to `confocal_random_zernike` as closely as `data/R80_N200_A1000000...f0.3000` would be (exact n_{atoms} /cloud-prior/shot-count match, already on disk, just never run through `generate_kinematic_estimates.py`) – worth doing if a tight, controlled aberration-only comparison is needed later.
- **Connect to the β (signal) CRB – done, §8–§8.4:** the aberrated dataset’s beta-CRB is 10^{-4} rad at $N=200$ (vs. $1e8$ ’s 1.8×10^{-5} at $N=50$) – aberration alone costs roughly a $6\times$ worse phase-information floor per shot even before any estimator-inefficiency gap is added on top. The general aberration-vs-Gaussian quantitative bound (the *original* motivating question) is now answerable in principle from what’s already computed here; a dedicated matched- N comparison (see below) would sharpen it.
- The current check used only `run_000`’s first 10 shots for the CRB side (cheap: ~ 1 minute for all of §3–4 combined). Widening this to more shots/runs would tighten the CRB-side estimate too (currently it isn’t the limiting source of noise in the comparison, but worth checking as the empirical run finishes all 20 runs).

- **Quantify the PSF-variance term for moments (§6.4):** the width-recovery failure of `moments` was diagnosed *qualitatively* (the ballistic-expansion matrix A has no PSF term, so $M = A\eta$ is itself wrong at the PSF scale) but not *quantitatively* – computing the PSF’s actual contribution to V_{xf} (from the PSMAP/optics forward model already used elsewhere in this repo) and adding it as a third bias term alongside Poisson noise (§6.3) and shrinkage bias (§6.4) should close the remaining $\sim 100\times$ gap for σ_x, σ_y and would make the `moments` error budget fully closed-form, not just qualitatively understood.
- **Explain why `moments- $\beta$` beats `best- $\beta$` for the aberrated dataset only (§8.4):** flagged as a real, counter-intuitive, unexplained result – the working hypothesis (that `best`’s two-stage cross-term loss interacts worst exactly where `theta`’s ridge is strongest) is plausible but untested; would need a direct comparison of the two estimators’ η -space error vectors on the same shots to confirm.
- **The systematic-bias-vs-noise-floor caveat (§9.3) is a general principle, not just a phase-shear one** – worth stating explicitly: *any* CRB in this note only models the noise floor of a correctly-specified model; a large raw empirical/CRB ratio anywhere in this document (e.g. the aberrated dataset’s β ratios of $22\text{--}29\times$, §8.4) should always be interrogated for a hidden systematic/model-misspecification component before being attributed purely to estimator inefficiency – the phase-shear section is the one place in this note where that check was actually done end-to-end (via the published fitted-feature/oracle corrections) and it turned out to matter enormously ($33\times \rightarrow 1.6\times$ once corrected). The kinematic pipeline’s own large gaps (aberrated β , `1e8` `theta` widths) have *not* yet been checked for an analogous hidden systematic – currently attributed to estimator inefficiency and non-convergence respectively, but neither has been ruled out as partly systematic in the same way phase-shear’s raw number was.